1.7 Effects of Local Masses on the Gravity and the Geoid: Illustration Using a Point Mass

We use a simple scenario, a point mass located nearby, to illustrate how local masses, such as mountains or subsurface minerals, affect the gravity and the geoid.

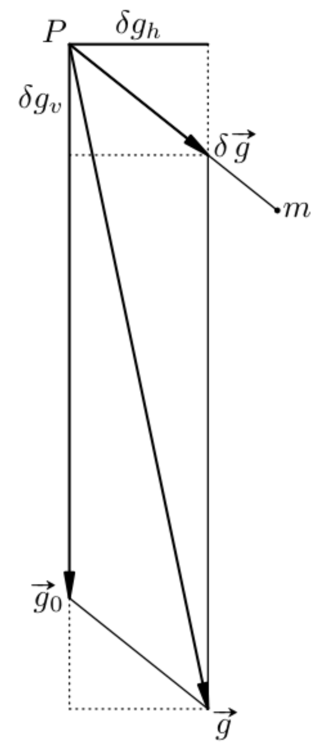
1.7.1 Effect on the gravity

As shown in Fig. 1.18, consider the gravity at a point P, where $\vec{g}_0$ is the gravity without the effect of the point mass m, $\delta\vec{g}$ is the gravitational attraction of m, and $\vec{g}$ is the total gravity. We denote the magnitudes of $\vec{g}_0$, $\delta\vec{g}$, and $\vec{g}$ as g_0, δg, and g, respectively. We further denote the component of $\delta\vec{g}$ along the direction of $\vec{g}_0$ as δg_v and that along the direction perpendicular to $\vec{g}_0$ as δg_h. We then have

$$g = \left[\left(g_0 + \delta g_v \right)^2 + \delta g_h^2 \right]^{1/2} . \tag{1.181}$$

We assume that the order of magnitude of δg is far smaller than that of g_0. We can then develop (1.181) as, taking into account of $\delta g = (\delta g_v^2 + \delta g_h^2)^{1/2}$ and neglecting the second and higher terms of $\delta g / g_0$,

Fig. 1.18 Effect of a nearby point mass on the gravity



$$g = \left(g_0{}^2 + 2g_0\delta g_v + \delta g_v^2 + \delta g_h^2\right)^{1/2} = g_0\left[1 + 2\frac{\delta g_v}{g_0} + \left(\frac{\delta g}{g_0}\right)^2\right]^{1/2}$$
$$\approx g_0\left(1 + \frac{\delta g_v}{g_0}\right) = g_0 + \delta g_v\,. \tag{1.182}$$

The angle between $\vec{g}_0$ and $\vec{g}$ is

$$(\vec{g}_0, \vec{g}) = \frac{\delta g_h}{g} \approx \frac{\delta g_h}{g_0}\,, \tag{1.183}$$

which is extremely small. Hence, we can approximately understand δg_v as the vertical component of $\delta\vec{g}$, i.e., the component along the total gravity $\vec{g}$. Consequently, we can approximately understand δg_h as the horizontal component of $\delta\vec{g}$.

In Fig. 1.18, the point mass m is assumed to be positive and is drawn below the horizontal plane through P for convenience, so that the downward vertical component δg_v is positive. The formulation applies equally to the case when the point mass m is negative or located above the horizontal place through P. These scenarios are reflected in the signs of the components δg_h and δg_v.

The problem formulated above agrees with many geophysical problems such as the gravity reduction to be formulated in Chap.6 and the exploration of subsurface materials using gravity data.

1.7.2 Effect on the geoid

Take reference to Fig. 1.19. We denote the geoid without the effect of the point mass m as Σ_0 and the geoid with the effect of m as Σ. We further denote the height of Σ above Σ_0 as δN, which is to be formulated.

We denote the gravity potential of the Earth excluding m as W_0, the gravitational potential of m as δV, and the gravity potential of the Earth including m as W. By definition, W_0 is constant on Σ_0, W is constant on Σ, and

$$W = W_0 + \delta V\,. \tag{1.184}$$

We need the difference between the values of W_0 on Σ and Σ_0. By definition, the gravity of the Earth excluding m on Σ_0 is, according to (1.48),

$$g_0(\Sigma_0) = -\left(\frac{\partial W_0}{\partial H}\right)_{H=0}\,, \tag{1.185}$$

where H is defined as the height above Σ_0. We can thus express the difference between the values of W_0 on Σ and Σ_0 as

$$\delta W_0(\delta N) = W_0(\Sigma) - W_0(\Sigma_0) = \left(\frac{\partial W_0}{\partial H}\right)_{H=0}\delta N = -g_0(\Sigma_0)\delta N\,. \tag{1.186}$$

This is a differential, which requires that both W_0 and $\partial W_0/\partial H$ are continuous in the vicinity of Σ_0. The Earth is a solid body and its gravity field does have such properties. We know that $g_0(\Sigma_0)$ is position dependent on Σ_0; therefore, $\delta W_0(\delta N)$ is position dependent.

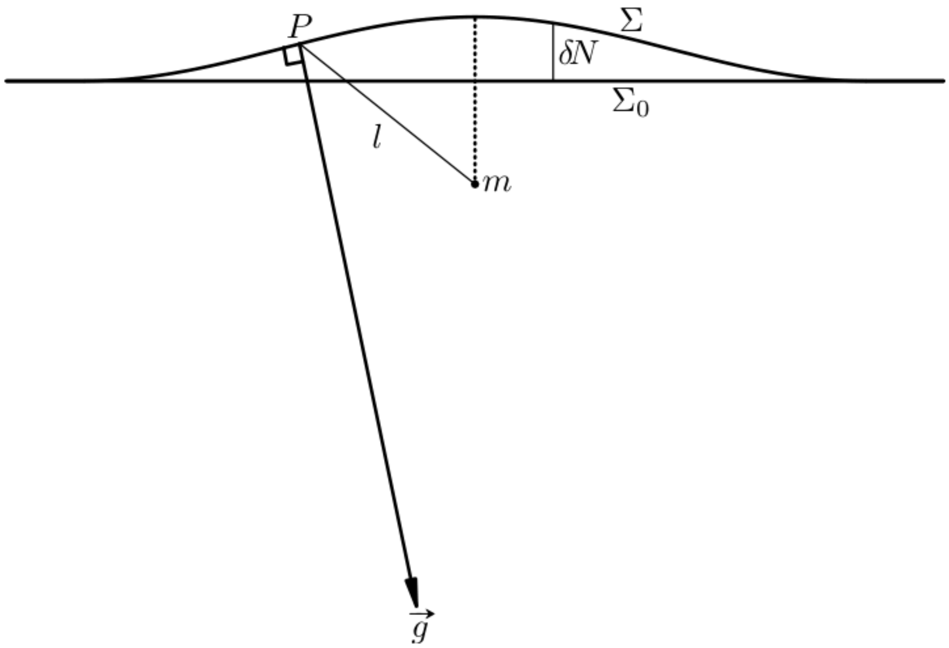


Fig. 1.19 Effect of a nearby point mass on the geoid

On Σ, we have, with (1.186),

$$\begin{aligned} W(\Sigma) &= W_0(\Sigma) + \delta V(\Sigma) = W_0(\Sigma_0) + \delta W_0(\delta N) + \delta V(\Sigma) \\ &= W_0(\Sigma_0) - g_0(\Sigma_0)\delta N + \delta V(\Sigma). \end{aligned} \tag{1.187}$$

By definition,

$$\delta V(\Sigma) = G\frac{m}{l}\,, \tag{1.188}$$

where l is the distance from m, such as that shown in Fig. 1.19 for the point P. The magnitude of δV is significant only when l is small, i.e., in the close proximity of m. Consequently, δN is significantly different from 0 only in the close proximity of m, also as shown in Fig. 1.19. Therefore, by applying (1.187) and (1.188) to a place very far from m, we obtain $\delta N = 0$ and $\delta V(\Sigma) = 0$ to a very good approximation, resulting in $W(\Sigma) = W_0(\Sigma)$. Hence, we have $-g_0(\Sigma_0)\delta N + \delta V(\Sigma) = 0$ based on (1.187), leading to

$$\delta N = \frac{\delta V(\Sigma)}{g_0(\Sigma_0)}\,. \tag{1.189}$$

If $m > 0$, δN is positive and the shape of Σ bulges upward above m, which reflects the fact that $\delta V(\Sigma)$ is inverse proportional to the distance from m, l, as shown in Fig. 1.19. This fact can also be explained based on the direction of the gravity: the gravitational attraction of m contributes to the total gravity on Σ, $\vec{g}(\Sigma)$, with a horizontal component pointing to the side of m. As the gravity is perpendicular to Σ, Σ must have a upward slope toward m. This fact is also shown in Fig. 1.19, which is based on Fig. 1.18.

If $m > 0$ and is located above, the vertical component of its gravitational attraction δg_v is negative, which makes g smaller than g_0 (imagine m is above the horizontal plane in Fig. 1.18). However, the horizontal component δg_h still points to the side of m. Hence, the shape of Σ is similar to that when m is below the horizon as shown in Fig. 1.19. This fact is straightforward to understand based on (1.189). The magnitude of gravity g does not contradict the shape of Σ. On an equipotential surface of gravity, according to (1.48), the gravity is inverse proportional to the distance to an adjacent equipotential surface of gravity, but not the shape of this surface.